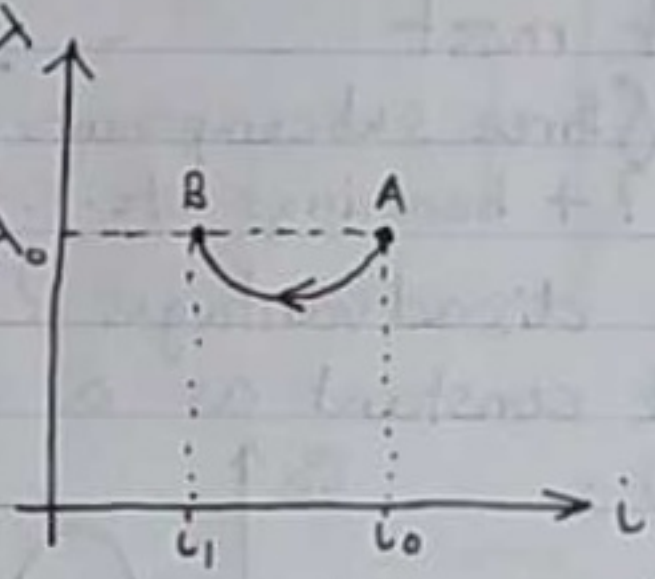
Lecture 27

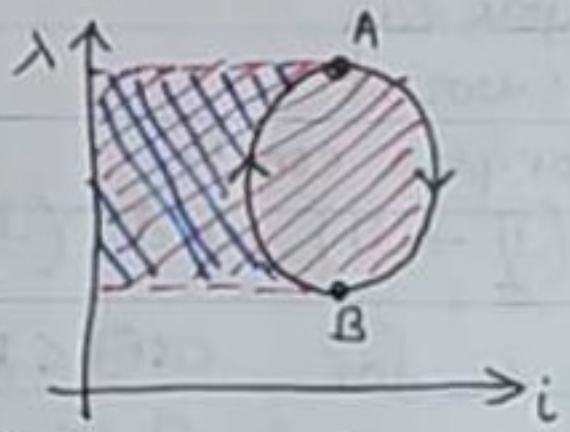
Last lecture : * Operation of a variable reluctance machine as generator
* More EFE, EFM, energy stored.
example.

Q.



EFE = ?
A→B
(a) Zero (b) Non-zero

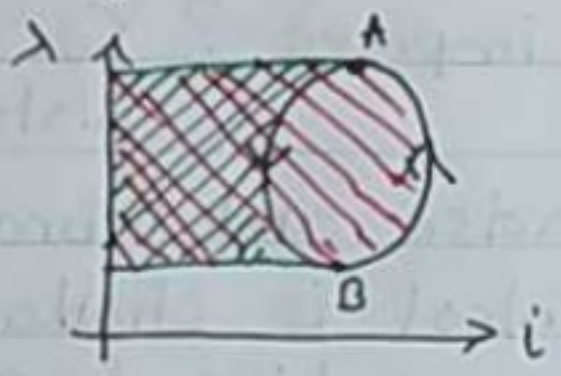
In general,



EFE = negative
A→B

EFE = positive
B→A

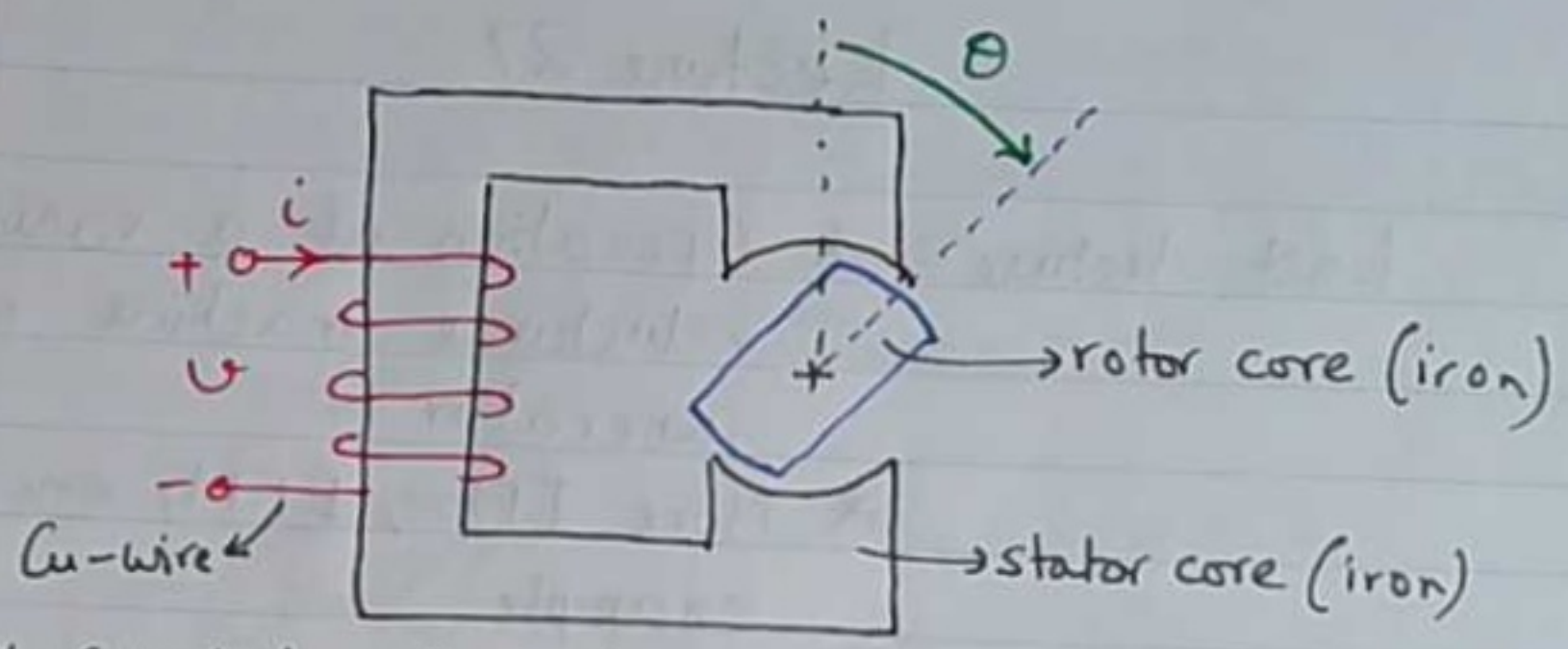
∮EFE < 0 (Generator)



EFE = negative
A→B

EFE = positive
B→A

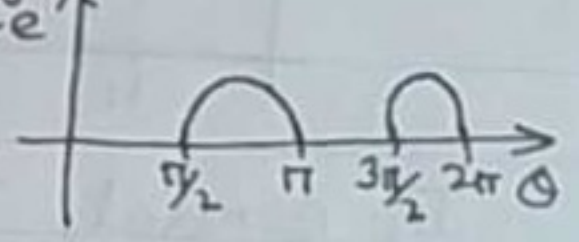
∮EFE > 0 (Motor).



* World's simplest most
- highly reliable {three subcomponents + bearings etc.} inexpensive,

Q. What are the disadvantages ?

* Torque is not constant as a function of θ. (probably you can change current to hold torque const. between $(\frac{\pi}{2} \to \pi)$ and $(\frac{3\pi}{2} \to 2\pi)$.



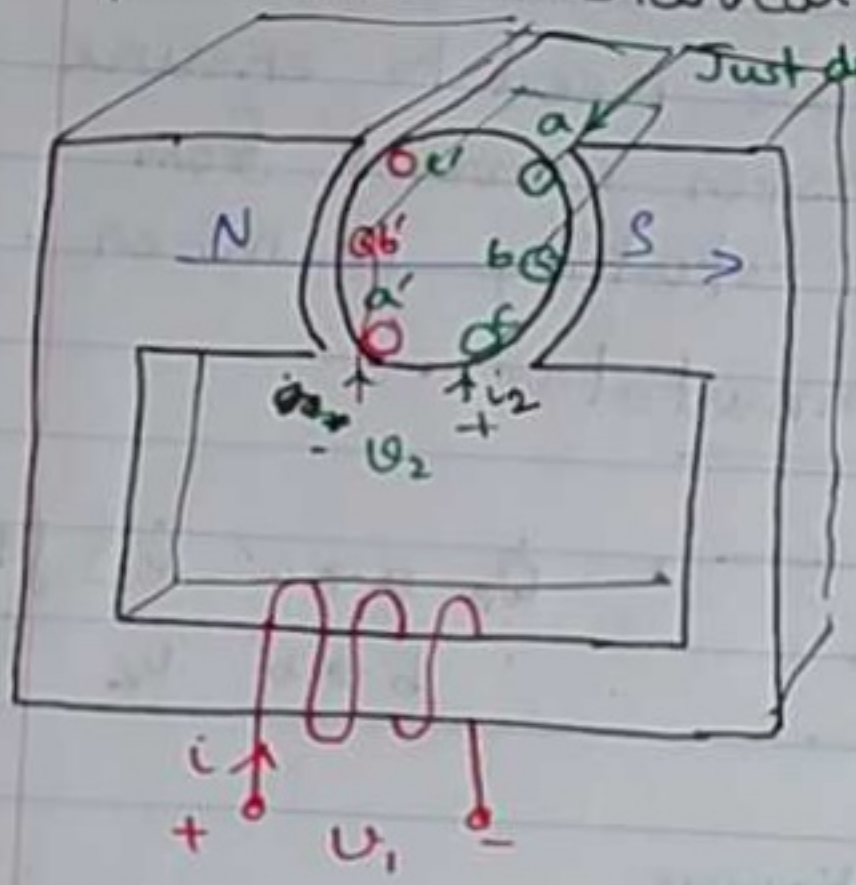
But $\tau = 0$, for $0 < \theta \leq \frac{\pi}{2}$ / $\pi < \theta < \frac{3\pi}{2}$

- Can you improve ? Yes add more stator segments.

* Acoustic noise and vibration
* Torque control is challenging. - highly non-linear machine. [coz inductance function is an approximation).

Q <u>Any alternative</u> ?

Q. Can we get continuous rotation ?
Let us go back to our two-electrical port-one-mechanical port example.



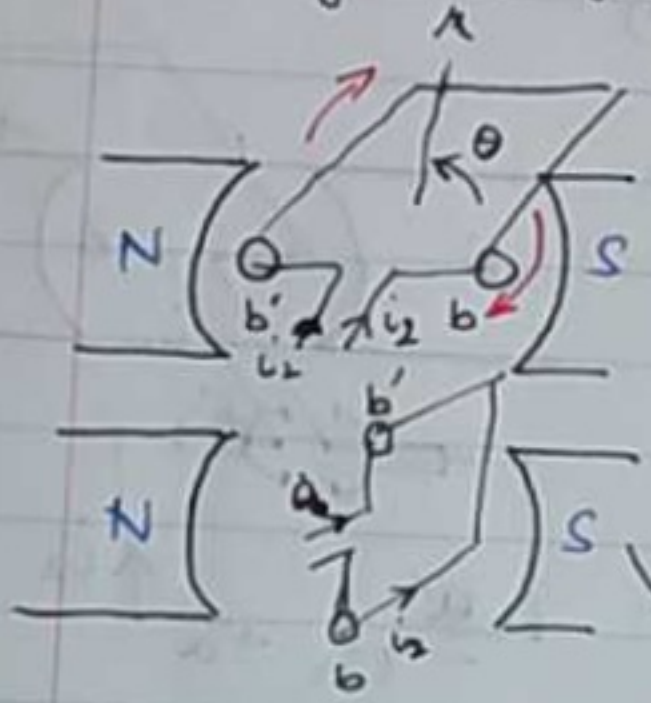
$$\lambda_1 = L_1 i_1 + M(\theta)\, i_2$$

$$\lambda_2 = M(\theta)\, i_1 + L_2 i_2$$

* Uniform airgap.

$$\tau_e = i_1 i_2 \frac{\partial M(\theta)}{\partial \theta}.$$

So if we assume $M(\theta) = K \cos\theta$.



$$\tau_e = -i_1 i_2 K \sin\theta.$$

$M(90°) = 0$ [Makes sense coil 2 and 1 are orthogonal]

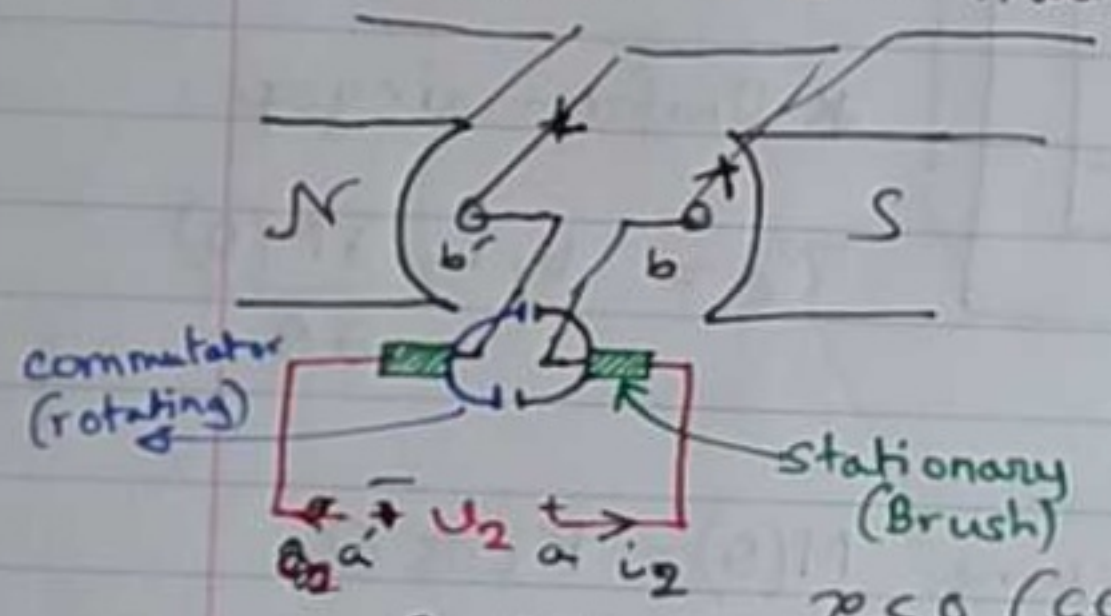
$\tau(90°) = -i_1 i_2 K$.

$M(0°) = K$.

$\tau(0°) = 0$.

Now what? If you keep i_1 and i_2 polarity same once $\theta < 0°$ τ_e polarity changes.
So, wdg bb' will remain aligned.

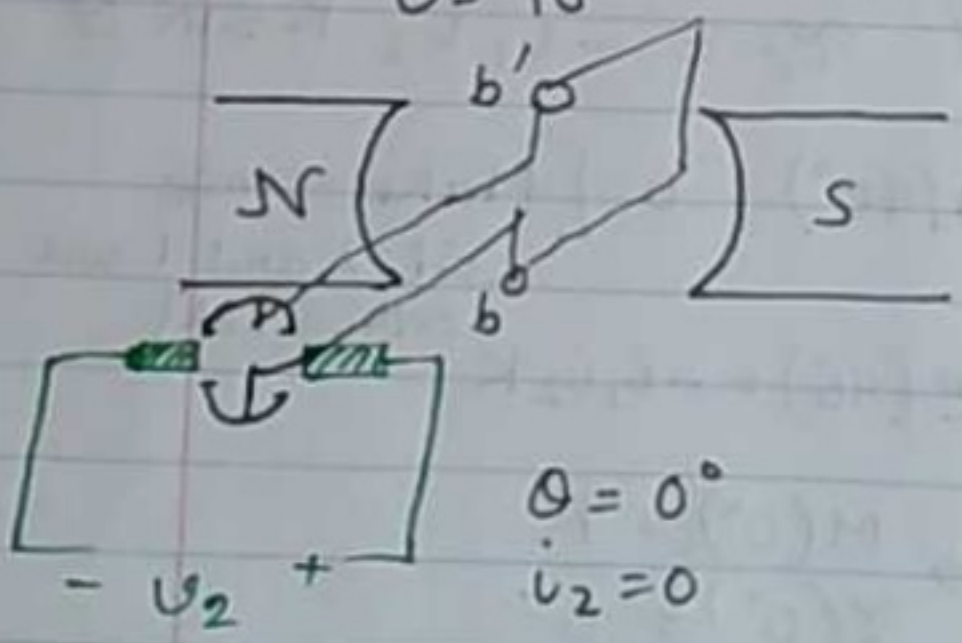
* Somehow we need to change one polarity either i_1 or i_2.
* Also, somehow, we need to ensure that we can transfer current from a stationary to rotating member.
  - Brush + commutator.



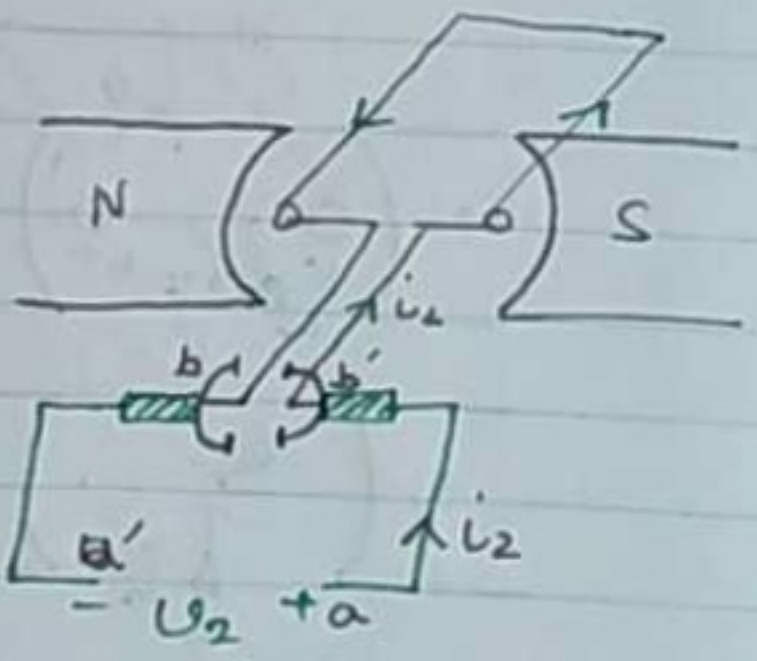
a → a' i_2 flows
b → b' i_2 flows.

$\tau_e < 0$ (cw)

$\theta = 90°$



$\theta = 0°$
$i_2 = 0$



a → a' i_2 flows
b' → b i_2 flow
$i_{bb'} < 0$

Brushed dc motor

$\tau_e < 0$ (ccw).

Why dc ?

→ the polarity of voltage, at terminal aa' is always positive.
— a dc supply is good enough.